Advanced Functions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Inverse functions with domain restrictions**

Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.

$f^{-1}(x) = \sqrt{x}$. The restriction $x \geq 0$ is necessary because $f(x) = x^2$ is not one-to-one over all real numbers (e.g. $f(2) = f(-2) = 4$), so it has no inverse function unless the domain is restricted to make it one-to-one.

2 **[Foundational]** • **Target: End behavior of polynomial functions**

Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.

Since the leading term is $-2x^4$ (even degree, negative coefficient), both ends of the graph point downward: as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

3 **[Foundational]** • **Target: End behavior of polynomial functions**

Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.

Since the leading term is $3x^5$ (odd degree, positive coefficient): as $x \to \infty$, $g(x) \to \infty$; as $x \to -\infty$, $g(x) \to -\infty$.

4 **[Foundational]** • **Target: Rational functions: asymptotes and holes**

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.

Vertical asymptote: where the denominator is zero: $x = 3$. Horizontal asymptote: since numerator and denominator have the same degree, the asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

5 **[Foundational]** • **Target: Even, odd, and periodic function classification**

Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.

$f(-x) = (-x)^4 - 3(-x)^2 = x^4 - 3x^2 = f(x)$. Since $f(-x) = f(x)$, the function is even.

6 **[Foundational]** • **Target: Finding zeros of higher-degree functions graphically and algebraically**

A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u-1)(u-4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

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7 [Foundational] • Target: Symmetry and transformations of rational functions

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

$f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$. Since $f(-x) = -f(x)$, the function is odd, meaning its graph has point symmetry about the origin — rotating the graph 180° about the origin maps it onto itself.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Finding a function's formula from a description of transformations

The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.

Shifting right by 2 replaces x with $x - 2$: $\sqrt{x-2}$. Vertical stretch by 3: $3\sqrt{x-2}$. Shift down 1: $y = 3\sqrt{x-2} - 1$.

9 [Intermediate] • Target: A multi-part rational function analysis problem

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.

(a) Numerator: $2x^2 - 8 = 2(x^2 - 4) = 2(x-2)(x+2)$. Denominator: $x^2 - x - 6 = (x-3)(x+2)$. (b) The factor $(x+2)$ cancels, indicating a hole at $x = -2$ (specifically at the point found by substituting into the simplified expression). The remaining denominator factor $(x-3)$ gives a vertical asymptote at $x = 3$ (not canceled). (c) Since the numerator and denominator have the same degree (both degree 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

10 [Intermediate] • Target: Finding a quadratic function from a table of values

A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a, b, and c.

From $f(0) = 5$: $c = 5$. From $f(1) = 10$: $a + b + c = 10$, so $a + b = 5$. From $f(2) = 19$: $4a + 2b + c = 19$, so $4a + 2b = 14$, i.e. $2a + b = 7$. Subtracting $a + b = 5$ from $2a + b = 7$: $a = 2$, then $b = 5 - 2 = 3$. Function: $f(x) = 2x^2 + 3x + 5$.

11 [Intermediate] • Target: Determining if a function has an inverse using the horizontal line test

Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.

The horizontal line test states that a function has an inverse (is one-to-one) if and only if no horizontal line intersects its graph more than once. For $f(x) = x^3$, every horizontal line $y = k$ intersects the graph exactly once (since every real number has exactly one real cube root), so $f(x) = x^3$ does have an inverse function, namely $f^{-1}(x) = \sqrt[3]{x}$.

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12 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \to \infty$ and $x \to -\infty$.

Since the leading term is $-2x^4$ (even degree, negative coefficient), both ends of the graph point downward: as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the vertical and horizontal asymptotes of $f(x) = \frac{2x+1}{x-3}$.

Vertical asymptote: where the denominator is zero: $x = 3$. Horizontal asymptote: since numerator and denominator have the same degree, the asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.

$f(-x) = \frac{1}{-x} = -\frac{1}{x} = -f(x)$. Since $f(-x) = -f(x)$, the function is odd, meaning its graph has point symmetry about the origin — rotating the graph 180° about the origin maps it onto itself.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Advanced Trigonometry

This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.

(a) Numerator: $2x^2 - 8 = 2(x^2 - 4) = 2(x-2)(x+2)$. Denominator: $x^2 - x - 6 = (x-3)(x+2)$. (b) The factor $(x+2)$ cancels, indicating a hole at $x = -2$ (specifically at the point found by substituting into the simplified expression). The remaining denominator factor $(x-3)$ gives a vertical asymptote at $x = 3$ (not canceled). (c) Since the numerator and denominator have the same degree (both degree 2), the horizontal asymptote is the ratio of leading coefficients: $y = \frac{2}{1} = 2$.

16 [Challenging] • Target: Composite functions with restrictions • Cross-Lesson: 5.01 Pre-Calculus Concepts

If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.

$(f \circ g)(x) = \sqrt{x-5}$, which requires $x - 5 \geq 0$, so the domain is $x \geq 5$.

17 [Challenging] • Target: Composite functions with restrictions • Cross-Lesson: 6.01 Advanced Coordinate Geometry

If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

$(f \circ g)(x) = f\left(\frac{1}{x}\right) = \left(\frac{1}{x}\right)^2 - 1 = \frac{1}{x^2} - 1$.

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18 [Challenging] • Target: Rational functions: asymptotes and holes • Cross-Lesson: 7.01 Advanced Plane Geometry

Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2 - 4}{x - 2}$.

Factor: $f(x) = \frac{(x-2)(x+2)}{x-2} = x + 2$ for $x \neq 2$. There is a hole at $x = 2$, at the point $(2, 4)$ (found by substituting $x = 2$ into the simplified expression $x + 2$).

19 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 8.01 Statistics, Counting & Probability

Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.

$\log_3(9) + \log_3(27) = \log_3(9 \times 27) = \log_3(243) = \log_3(3^5) = 5$.

20 [Challenging] • Target: Logarithmic functions and their properties • Cross-Lesson: 1.01 Advanced Number Theory

Solve $\ln(x) + \ln(x - 3) = \ln(10)$.

Combine logs: $\ln[x(x-3)] = \ln(10)$, so $x(x-3) = 10$, giving $x^2 - 3x - 10 = 0$. Factor: $(x-5)(x+2) = 0$, so $x = 5$ or $x = -2$. Since $\ln(x)$ requires $x > 0$, reject $x = -2$. Solution: $x = 5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Composite functions with restrictions

Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$

d. ∞, ∞ ✓

The leading term is $4x^4$ (even degree, positive coefficient), so as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$.

22 [Foundational] • Target: Inverse functions with domain restrictions

Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$

b. $-\infty, \infty$ ✓

c. ∞, ∞

d. $-\infty, -\infty$

The leading term is $-x^3$ (odd degree, negative coefficient), so as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$.

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23 [Foundational] • Target: End behavior of polynomial functions

Describe the end behavior of $f(x) = -4x^4 - 2x^2$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. ∞, ∞

d. $-\infty, -\infty$ ✓

The leading term is $-4x^4$ (even degree, negative coefficient), so as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

24 [Foundational] • Target: Rational functions: asymptotes and holes

Describe the end behavior of $f(x) = x^2 - 2x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. ∞, ∞ ✓

b. $-\infty, -\infty$

c. $-\infty, \infty$

d. $\infty, -\infty$

The leading term is x^2 (even degree, positive coefficient), so as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to \infty$.

25 [Foundational] • Target: Logarithmic functions and their properties

Describe the end behavior of $f(x) = -4x^2 + 4x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. ∞, ∞

b. $-\infty, -\infty$ ✓

c. $\infty, -\infty$

d. $-\infty, \infty$

The leading term is $-4x^2$ (even degree, negative coefficient), so as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

26 [Foundational] • Target: Even, odd, and periodic function classification

Describe the end behavior of $f(x) = x^5 - x^3$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$ ✓

b. $-\infty, \infty$

c. $-\infty, -\infty$

d. ∞, ∞

The leading term is x^5 (odd degree, positive coefficient), so as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$.

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27 [Foundational] • Target: Piecewise-defined functions and continuity

Find the vertical asymptote of $f(x) = \frac{x-15}{x+14}$.

a. -13 **b.** -15

c. 14 **d.** -14 ✓

The vertical asymptote occurs where the denominator is zero: $x + 14 = 0$, so $x = -14$.

28 [Foundational] • Target: A multi-part function transformation and composition problem

Find the vertical asymptote of $f(x) = \frac{x-7}{2x-12}$.

a. 7 **b.** -6

c. 6 ✓ **d.** 5

The vertical asymptote occurs where the denominator is zero: $2x - 12 = 0$, so $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Finding zeros of higher-degree functions graphically and algebraically

Find the vertical asymptote of $f(x) = \frac{-8x-1}{7x+7}$.

a. 1 **b.** -1 ✓

c. 0 **d.** -2

The vertical asymptote occurs where the denominator is zero: $7x + 7 = 0$, so $x = -1$.

30 [Intermediate] • Target: Function operations: sum, difference, product, quotient

Find the vertical asymptote of $f(x) = \frac{9x-10}{x-7}$.

a. -7 **b.** 7 ✓

c. 6 **d.** 8

The vertical asymptote occurs where the denominator is zero: $x - 7 = 0$, so $x = 7$.

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31 [Intermediate] • Target: Symmetry and transformations of rational functions

Find the vertical asymptote of $f(x) = \frac{5x+4}{-5x-15}$.

a. -4

b. 3

c. -3 ✓

d. -2

The vertical asymptote occurs where the denominator is zero: $-5x - 15 = 0$, so $x = -3$.

32 [Intermediate] • Target: Finding a function's formula from a description of transformations

Find the vertical asymptote of $f(x) = \frac{x-12}{7x-14}$.

a. -2

b. 2 ✓

c. 3

d. 1

The vertical asymptote occurs where the denominator is zero: $7x - 14 = 0$, so $x = 2$.

33 [Intermediate] • Target: A multi-part rational function analysis problem

Solve $\ln(x) + \ln(x-5) = \ln(55)$.

a. 11 ✓

b. 5

c. 10

d. 12

Combine logs: $\ln[x(x-5)] = \ln(55)$, so $x(x-5) = 55$, giving $x^2 - 5x - 55 = 0$. Factor: $(x-11)(x-(-5)) = 0$, so $x = 11$ or $x = -5$. Since $\ln(x)$ requires $x > 0$, reject $x = -5$. Solution: $x = 11$.

34 [Intermediate] • Target: Finding a quadratic function from a table of values

Solve $\ln(x) + \ln(x-8) = \ln(32)$.

a. 3

b. 8

c. 4 ✓

d. 5

Combine logs: $\ln[x(x-8)] = \ln(32)$, so $x(x-8) = 32$, giving $x^2 - 8x - 32 = 0$. Factor: $(x-4)(x-(-8)) = 0$, so $x = 4$ or $x = -8$. Since $\ln(x)$ requires $x > 0$, reject $x = -8$. Solution: $x = 4$.

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35 [Intermediate] • Target: Determining if a function has an inverse using the horizontal line test

The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-2, -1, 1, 2$ ✓

b. $-1, -2, 1, 2$

c. $1, 2$

d. $-2, 2$

Let $u = x^2$: $u^2 - 5u + 4 = 0$, factoring to $(u - 1)(u - 4) = 0$, so $u = 1$ or $u = 4$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 4$ gives $x = \pm 2$. All real zeros: $x = -2, -1, 1, 2$.

36 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The quartic function $f(x) = x^4 - 17x^2 + 16$ can be treated as a quadratic in x^2. Find all real zeros.

a. $-1, -4, 1, 4$

b. $1, 4$

c. $-4, -1, 1, 4$ ✓

d. $-4, 4$

Let $u = x^2$: $u^2 - 17u + 16 = 0$, factoring to $(u - 1)(u - 16) = 0$, so $u = 1$ or $u = 16$. Since $u = x^2$: $x^2 = 1$ gives $x = \pm 1$; $x^2 = 16$ gives $x = \pm 4$. All real zeros: $x = -4, -1, 1, 4$.

37 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

If $f(x) = -3x + 5$ and $g(x) = x^2 + 5$, find $(f \circ g)(3)$.

a. -39

b. -37 ✓

c. -35

d. 14

First, $g(3) = (3)^2 + 5 = 14$. Then $f(14) = -3(14) + 5 = -37$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $f(x) = 2x - 1$ and $g(x) = -2x^2$, find $(f \circ g)(-1)$.

a. -7

b. -2

c. -3

d. -5 ✓

First, $g(-1) = -2(-1)^2 = -2$. Then $f(-2) = 2(-2) - 1 = -5$.

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39 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 8.01 Statistics, Counting & Probability

Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $\infty, -\infty$

b. $-\infty, \infty$ ✓

c. ∞, ∞

d. $-\infty, -\infty$

The leading term is $-x^3$ (odd degree, negative coefficient), so as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to \infty$.

40 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Advanced Number Theory

Describe the end behavior of $f(x) = 4x^5 + 5x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, -\infty$

b. $\infty, -\infty$ ✓

c. ∞, ∞

d. $-\infty, \infty$

The leading term is $4x^5$ (odd degree, positive coefficient), so as $x \to \infty$, $f(x) \to \infty$; as $x \to -\infty$, $f(x) \to -\infty$.

41 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Advanced Algebra & Systems

Describe the end behavior of $f(x) = -3x^2 - 9x$: as $x \to \infty$, $f(x) \to ?$, and as $x \to -\infty$, $f(x) \to ?$

a. $-\infty, \infty$

b. $\infty, -\infty$

c. $-\infty, -\infty$ ✓

d. ∞, ∞

The leading term is $-3x^2$ (even degree, negative coefficient), so as $x \to \infty$, $f(x) \to -\infty$; as $x \to -\infty$, $f(x) \to -\infty$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 4.01 Advanced Trigonometry

Find the vertical asymptote of $f(x) = \frac{2x+11}{-x-8}$.

a. -7

b. 8

c. -9

d. -8 ✓

The vertical asymptote occurs where the denominator is zero: $-x - 8 = 0$, so $x = -8$.

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43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.01 Pre-Calculus Concepts

Find the vertical asymptote of $f(x) = \frac{5x+4}{-5x-15}$.

a. -4

b. 3

c. -3 ✓

d. -2

The vertical asymptote occurs where the denominator is zero: $-5x - 15 = 0$, so $x = -3$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Find the vertical asymptote of $f(x) = \frac{7x+14}{3x-9}$.

a. 2

b. 4

c. -3

d. 3 ✓

The vertical asymptote occurs where the denominator is zero: $3x - 9 = 0$, so $x = 3$.

45 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\log_b 3 = x$ and $\log_b 5 = y$, what is $\log_b (45)$ in terms of x and y?

a. $2x + y$ ✓

b. $x^2 + y$

c. $2xy$

d. $x + 2y$

Factor $45 = 3^2 \times 5$. Using log properties: $\log_b (3^2 \times 5) = 2\log_b 3 + \log_b 5 = 2x + y$.

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